

OPEN ACCESS

Dosimetric characterization of a p-n silicon carbide detector under 320 kV X-ray irradiation

To cite this article: G. Petringa *et al* 2025 *JINST* **20** C08019

View the [article online](#) for updates and enhancements.

You may also like

- [Effect of Intermetallic Particle Size and Distribution on the Corrosion of an Mg-Al Alloy](#)
Leslie Gail Bland, Jishnu Bhattacharyya, Sean Agnew et al.
- [Formation of Metal Nanoparticles in Ionic Liquids By Low-Energy Quantum Beam Irradiation Technique](#)
Akihito Imanishi
- [Plasmonic Solar Fuels Based on Nanostructured Oxide Photocatalysts](#)
Dong Ha KIM, Yoon Hee Jang, Saji Thomas Kochuveedu et al.

INTERNATIONAL WORKSHOP ON
DETECTION SYSTEMS AND TECHNIQUES FOR FUNDAMENTAL AND APPLIED PHYSICS
CATANIA, ITALY
24–26 FEBRUARY 2025

Dosimetric characterization of a p-n silicon carbide detector under 320 kV X-ray irradiation

G. Petringa^{a,*}, A. Kurmanova,^a M. Guarrera,^a R. Catalano,^a C. Verona,^b L. Raffaele,^a
V. Salamone,^c S. Tudisco,^a G. Cuttone^a and G.A.P. Cirrone^a

^aNational Laboratories of the South, National Institute of Nuclear Physics,
Via Santa Sofia 62, Catania, Italy

^bIndustrial Engineering Department, “Tor Vergata” University of Rome,
Via Cracovia 50, Rome, Italy

^cDepartment of Medical Surgical Sciences and Advanced Technologies “G.F. Ingrassia”,
University of Catania,
Via Santa Sofia 78, Catania, Italy

E-mail: giada.petringa@lns.infn.it

ABSTRACT. This work presents the comprehensive dosimetric characterization of a p-n silicon carbide (SiC) detector exposed to a 320 kV_p X-ray beam, with the aim of evaluating its potential for preclinical and translational medical applications. The detector, based on a 10 µm thick 4H-SiC epitaxial layer, was assessed in terms of linearity with absorbed dose, dose rate dependence, percentage depth dose (PDD) distribution, response variation with field size, and short- and long-term reproducibility. Results demonstrated a highly linear dose-response relationship and negligible dose-rate dependence over the investigated range. The SiC device exhibited a consistent performance across different field sizes and penetration depths, highlighting its suitability for accurate dosimetry under medium-energy X-ray irradiation. Given its wide bandgap, low leakage current, high radiation tolerance, and near-tissue equivalence, the characterized SiC detector appears as a promising alternative to conventional dosimetric systems for radiobiological experiments and preclinical quality assurance in intermediate energy photon beams.

KEYWORDS: Dosimetry concepts and apparatus; Instrumentation for gamma-electron therapy

*Corresponding author.



Contents

1	Introduction	1
2	Material and methods	2
2.1	Detector description	2
2.2	Experimental setup	3
3	Results	4
3.1	Linearity with dose and dose rate dependence	4
3.2	Percentage Depth Dose distribution	5
3.3	Response variation with field size	6
3.4	Short-term and long-term reproducibility	7
4	Conclusion	7

1 Introduction

In the context of both preclinical research and translational medical physics, X-ray tubes operating at medium-to-high voltages (up to several hundred kilovolts) are widely employed for in-vitro cell irradiation, in-vivo animal studies, and specific diagnostic or therapeutic procedures such as intraoperative radiotherapy (IORT) and orthovoltage therapy. In particular, 320 kV_p X-ray beams represent a radiation beam quality of increasing relevance for radiobiological experiments aiming to investigate radiation effects under controlled conditions, where accurate dosimetric quantification is crucial. Moreover, such beams are frequently used for the calibration and benchmarking of detectors and phantoms under photon fields more penetrating than conventional diagnostic X-rays but less complex than MV beams from medical linear accelerators [1–3]. In these settings, the dosimeter must ensure high linearity across a wide dose range (from a few cGy to several Gy), stable response under repeated exposures (reproducibility), independence from the dose rate, and minimal sensitivity to ambient conditions such as temperature or humidity. Additionally, for both in-vitro and in-vivo applications, the detector geometry should be compact and tissue-equivalent to reduce perturbations of the irradiation field and ensure spatial accuracy of dose measure, especially when it must be delivered in embedded or adjacent configurations in cell cultures or in small animal setups. Conventional dosimeters such as ionization chambers, silicon diodes, and thermoluminescent dosimeters (TLDs) are often used in this context. However, they are not exempt from limitations. Ionization chambers, though considered reference-grade instruments, have relatively large volumes. Silicon diodes are compact and sensitive, but their performance degrades significantly with accumulated dose, exhibiting increased leakage currents and reduced signal stability [4, 5]. Recent technological developments in solid-state dosimetry have highlighted the potential of wide bandgap semiconductors, and among them, silicon carbide (SiC) has emerged as a promising candidate. SiC exhibits several advantageous properties for radiation detection: a wide bandgap (≈ 3.2 eV for 4H-SiC), extremely low leakage current at room temperature, high breakdown electric field (≈ 3 – 4 MV/cm), strong radiation hardness, and high thermal

conductivity [6]. These features enable the operation of SiC detectors in high-dose, high-gradient, and variable environmental conditions, making them attractive for both real-time and passive dosimetry. Studies have already demonstrated the suitability of SiC-based detectors in high-energy photon and electron beams, as well as proton and ion beams used in therapy or research contexts [7, 8]. In particular, their response has shown excellent linearity with dose, independence from dose rate, and negligible priming effects under prolonged irradiation, even without active cooling or frequent recalibration [9–12]. Moreover, unlike silicon, SiC has a lower effective atomic number ($Z_{\text{eff}} \approx 10$) closer to soft tissue ($Z_{\text{eff}} \approx 7.4$), reducing the over-response to low-energy photons typical of silicon-based detectors. Nevertheless, a systematic dosimetric evaluation of SiC detectors under medium-energy X-ray beams such as those produced by a 320 kV_p tube remains scarce in the literature. This beam quality is characterized by a complex spectral composition, with a significant contribution from photons between 100 and 300 keV, and necessitates dedicated calibration and response assessment, particularly when the detector is to be used in radiobiological protocols or quality assurance procedures. The aim of this work is to present a dosimetric characterization of a p-n junction SiC detector exposed to a 320 kV_p X-ray beam. The study includes measurements of the detector’s linearity with absorbed dose, dependence on dose rate, short-term and long-term reproducibility, percentage depth dose (PDD) behavior in a water-equivalent phantom, and response variation with field size (field factor). These parameters are essential to validate the performance of the device for its potential use in in-vitro cell irradiation, in-vivo small animal studies, and preclinical QA (Quality Assurance) procedures where compactness, stability, and energy independence are critical. By benchmarking the performance of the SiC detector against the typical requirements for radiobiological and clinical dosimetry, this work aims to support the deployment of SiC technology as a robust and versatile alternative to conventional dosimetric systems, particularly in intermediate-energy photon fields that are currently underserved by most commercial detectors.

2 Material and methods

2.1 Detector description

The dosimetric system investigated in this study is based on a novel solid-state detector fabricated using 4H-Silicon Carbide (4H-SiC) technology. The device features a p-n junction architecture, specifically optimized for radiation dosimetry. The sensitive volume consists of a 10 μm thick n-type epitaxial layer with a doping concentration on the order of $0.5\text{--}1 \times 10^{14} \text{ cm}^{-3}$, grown on top of a heavily doped ($N_A \approx 10^{19} \text{ cm}^{-3}$) p+ substrate. This junction geometry ensures efficient charge collection and high electric field uniformity across the active volume under reverse bias conditions. Electrical characterization performed prior to irradiation demonstrated full depletion at voltages around 10 V, with operational stability maintained under overdepletion conditions up to 30 V. Such biasing ensures maximal drift velocity and optimal charge collection efficiency, even under rapid or pulsed irradiation scenarios [7]. The active area of the detector is 4 mm² (2×2 mm²), which makes it well-suited for applications in preclinical dosimetry, especially where lateral field uniformity and spatial resolution are important. The device is mounted on a dedicated printed circuit board (PCB). For practical use in water-equivalent conditions, the entire assembly is enclosed in a cylindrical metallic support and encapsulated in black epoxy resin, allowing the detector to be safely immersed within a water phantom during measurements. This configuration ensures the mechanical robustness of the device while minimizing perturbations to the radiation field (see figure 1).

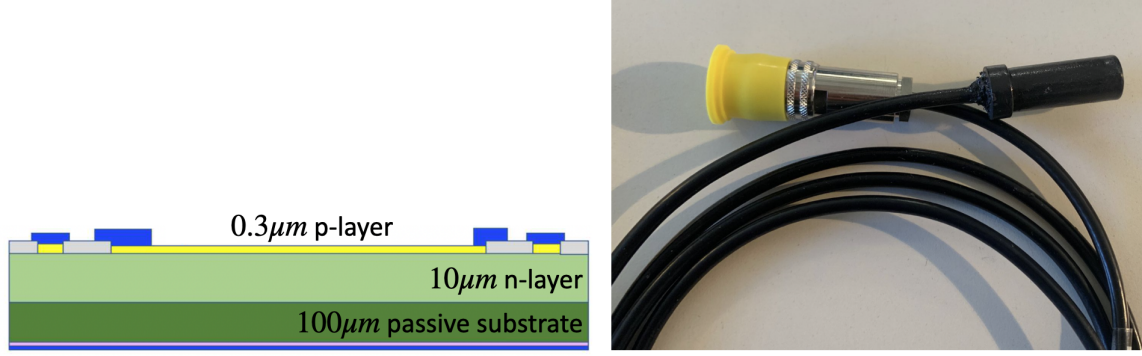


Figure 1. Left: schematic representation of the SiC detector architecture. The device features a planar p-n junction structure, composed of a $0.3\ \mu\text{m}$ heavily doped p^+ layer ($N_A \approx 10^{19}\ \text{cm}^{-3}$) and a $10\ \mu\text{m}$ low-doped n-type epitaxial layer ($0.5\text{--}1 \times 10^{14}\ \text{cm}^{-3}$), grown on a conductive n^+ SiC substrate. Right: photograph of the encapsulated SiC detector, mounted on a PCB, enclosed in a cylindrical metallic holder, and coated with black epoxy resin.

Electrical contacts to the diode were realized via wire bonding, ensuring low-impedance connections to the readout electronics. The detector exhibits extremely low leakage currents, even at elevated reverse bias voltages, a characteristic attributed to the wide bandgap nature of SiC ($\approx 3.26\ \text{eV}$) and the high crystalline quality of the epitaxial layers. This enables precise current measurements in the picoampere range, essential for dosimetry at low-to-moderate dose rates.

2.2 Experimental setup

The dosimetric characterization of the SiC detector was carried out using a preclinical X-ray irradiation system operating at 320 kV, based on a modularly adapted industrial tube (mod. HPX-320-11, Varex Imaging, U.S.A.). The source consists of a tungsten target with a 11° anode angle and a focal spot of 1 mm, generating a beam filtered by 4 mm beryllium to achieve a hardened spectrum. The beam collimation system includes an aperture ranging from 0 to 150 mm. All measurements were performed with the detector operated under a reverse bias voltage of 30 V ensuring complete depletion and maximal charge collection. In agreement with international dosimetric protocols for kilovoltage photon beams, the detector was positioned at the water surface for measurements of dose linearity, reproducibility, and dose rate dependence (see figure 2). For the percentage depth dose measurements, the detector was translated along the beam axis (z-direction) using a micrometric stage, with a positioning accuracy of 0.1 mm.

This setup allowed the acquisition of high-resolution depth dose profiles under reference conditions. The correct alignment of the detector within the irradiation field was ensured using a visible light field projected along the beam axis, which replicates the irradiation geometry and enables accurate centering and reproducibility. To ensure the accuracy of the absorbed dose delivered to the SiC detector, a cylindrical Farmer-type ionization chamber (model 30010) [13] was employed as a reference dosimeter. The reference chamber was positioned in the same geometric configuration as the SiC detector. A cross-calibration of the two systems allowed dose estimation uncertainties to be maintained below 3%. An Advanced Markus chamber (model 34045) [14] was also employed as a reference detector for the measurement of the Percentage Depth Dose (PDD) distribution in water and the field factor. This choice ensured high spatial resolution in the reference measurements. The detector signal was

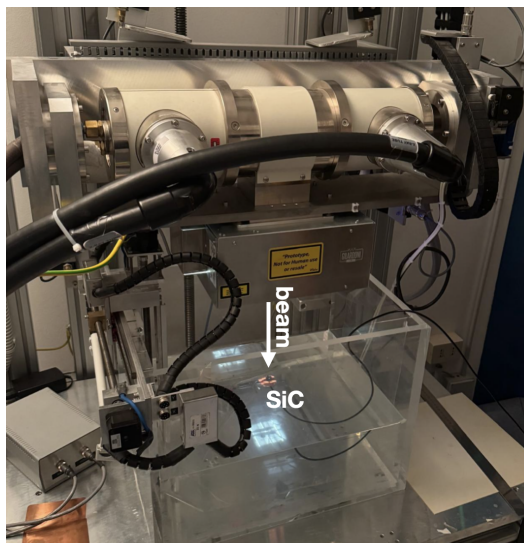


Figure 2. Schematic representation of the experimental setup used for dosimetric measurements. The SiC detector is positioned on the surface of the water inside a water phantom, aligned along the central axis of the photon beam. The X-ray tube, operating at 320 kV, is located at a fixed SSD distance of 50 cm from the water surface. The setup allows for reproducible detector positioning and ensures reference conditions for measurements of dose linearity, reproducibility, and dose rate dependence.

read out via low-noise coaxial cables connected to a Keithley 6517B electrometer [15]. The ambient conditions, including temperature and pressure, were monitored continuously, and corrections were applied accordingly to maintain the accuracy of the dosimetric data.

3 Results

3.1 Linearity with dose and dose rate dependence

The SiC detector demonstrated excellent linearity between the collected charge and the delivered dose, as illustrated in figure 3 (top). The detector was exposed to increasing dose levels, and for each point, the accumulated charge was measured. The resulting plot revealed a robust linear trend, quantitatively supported by the linear fit shown in the graph. Notably, the coefficient of determination R^2 derived from the fit is equal to 1, confirming the perfect correlation between dose and response within the investigated range. This result underscores the exceptional linearity and reliability of the detector for quantitative dosimetry. The bottom panel of figure 3 further reinforces this finding, showing that the sensitivity (charge per unit dose) remained consistent across all tested dose values. Fluctuations observed at lower doses are negligible and can be attributed to statistical variations. The low standard deviation of the data, indicated by the error bars, confirms the high repeatability of the measurements. This behavior underscores the detector's capability for dose quantification in applications where a broad dynamic range and high linearity are paramount.

Further validation was provided by the dose rate dependence test shown in figure 4. Here, the detector was irradiated at a constant dose (2 Gy) but with varying dose rates. The measured current response (nA) was plotted as a function of the dose rate, revealing a nearly linear profile.

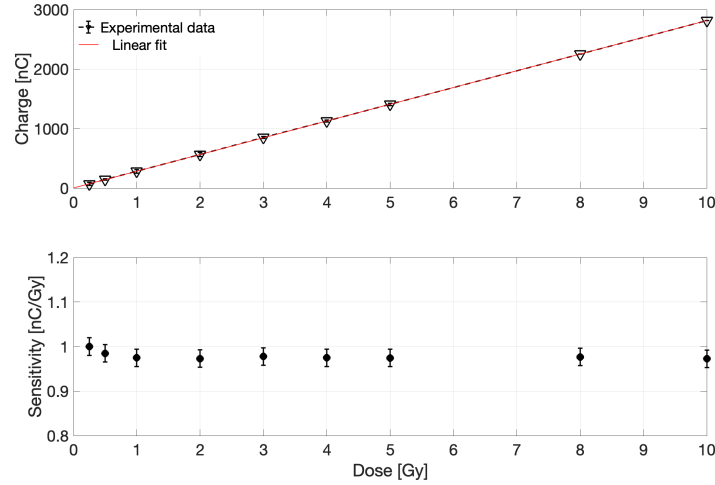


Figure 3. Top: charge collected by the SiC detector as a function of the delivered dose. Bottom: sensitivity of the SiC detector as a function of the delivered dose. Error bars represent the standard deviation of the experimental data corresponding to each dose point.

This affirms the detector’s dose rate independence — a critical requirement for accurate dosimetry in both conventional and high-intensity pulsed regimes. The stability of the current signal across a wide range of dose rates confirms the SiC detector’s robustness for real-time monitoring, even under rapidly varying irradiation conditions.

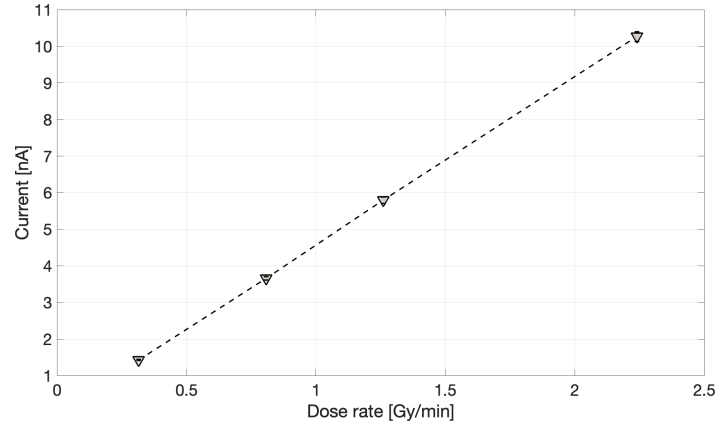


Figure 4. Current response of the SiC detector, expressed in nanoamperes (nA), as a function of the dose rate. The delivered dose was kept constant at 2 Gy. This analysis evaluates the dependence of the detector signal on the dose rate.

3.2 Percentage Depth Dose distribution

The depth-dose behavior of the SiC detector was evaluated in a water-equivalent phantom, with measurements performed at various depths along the beam axis. As depicted in figure 5, the Percentage Depth Dose curve exhibits the expected exponential attenuation profile characteristic of kilovoltage X-ray beams. The curve was normalized to the surface dose and provides insight into beam penetration and energy deposition within the phantom. The smoothness and agreement of the PDD profile

with theoretical expectations confirm not only the detector's spatial resolution and signal stability, but also its ability to accurately follow the variation in photon energy spectrum with depth. This is a nontrivial result: as X-ray beams undergo spectral hardening in matter, a detector affected by energy dependence would manifest distortions in the reconstructed PDD. The fact that the SiC detector is able to reproduce the correct dose-depth curve indicates that its response is effectively independent of the photon energy in the range investigated (up to 320 kV). This property is particularly advantageous in clinical and preclinical dosimetry where energy fluctuations are unavoidable, and simplifies the calibration procedures and correction models otherwise required for conventional detectors. Thus, the performance demonstrated in this context strengthens the role of SiC detectors as precise tools for beam characterization in translational and radiobiological studies requiring detailed knowledge of dose distribution.

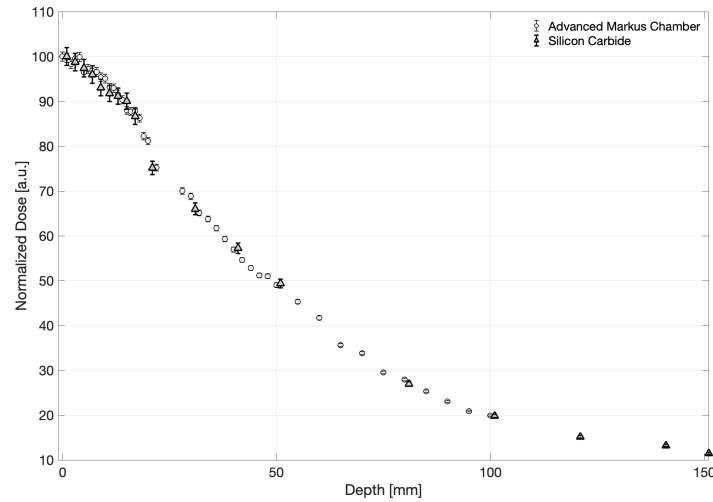


Figure 5. Percentage Depth Dose (PDD) distribution measured with the SiC detector. The dose values are normalized to the dose at the water surface (entrance) and plotted as a function of depth in water. The PDD acquired with the Advanced Markus Chamber is also shown as a reference.

3.3 Response variation with field size

Field size dependence was investigated by varying the radiation aperture while maintaining constant irradiation conditions. As shown in figure 6, the Relative Dose Factor (RDF) is plotted for both the SiC detector and the Advanced Markus chamber used as a reference. The RDF is defined as the ratio of the dose measured at a given field size to that measured at a reference field size, and it provides a normalized way to assess how the detector response varies with field geometry. For both detectors, the RDF exhibits a monotonic increase with field size, as expected for X-ray tube irradiation where larger apertures expose a greater area to both primary and scattered radiation. The close agreement between the SiC detector and the Markus chamber confirms the ability of the SiC device to accurately capture field-dependent dose variations. The trend suggests that no saturation or significant non-uniformity occurs within the active volume of the SiC sensor. This confirms its suitability not only for absolute dose measurements but also for detecting geometric modulation effects, which is particularly valuable in beam characterization and small-field dosimetry.

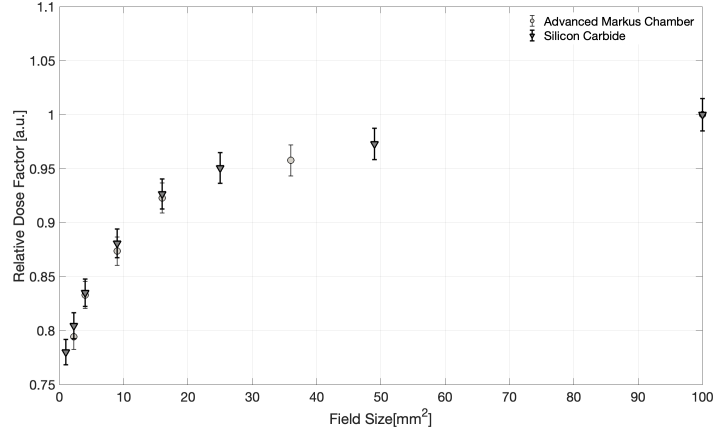


Figure 6. Variation of the charge collected by the SiC detector as a function of the radiation field size. The detector was positioned at the geometric center of the radiation field, and each measurement was performed under constant irradiation conditions: a delivered dose of 2 Gy and a dose rate of 0.8 Gy/min. The Advanced Markus chamber was used as a reference detector, and its response is also shown in the plot for comparison.

3.4 Short-term and long-term reproducibility

The reproducibility of the SiC detector was investigated through a series of repeated irradiations under fixed conditions. For the short-term evaluation, ten consecutive measurements were performed at a constant dose of 2 Gy and a dose rate of 0.8 Gy/min within a time window of 10 minutes. The maximum variation in the measured charge across the ten irradiations was found to be always below 1%, while the standard deviation remained within 0.1%, highlighting the excellent repeatability of the detector response in stable conditions. Long-term stability was assessed by monitoring the calibration factor of the detector over a period of 60 hours, maintaining the same irradiation parameters (2 Gy per exposure at 0.8 Gy/min). Throughout this duration, the maximum variation in detector sensitivity did not exceed 2%, confirming a high degree of stability over time. These results demonstrate that the SiC detector fulfills the key requirements for relative dosimetry in both short and extended timescales. Its high reproducibility, with minimal signal drift or fluctuation, ensures reliability for repeated use in preclinical and quality assurance procedures, without the need for frequent recalibration.

4 Conclusion

The dosimetric characterization of the p-n junction silicon carbide (SiC) detector under 320 kV X-ray irradiation has demonstrated its suitability for precision preclinical applications. The detector exhibited a highly linear response with absorbed dose ($R^2 > 0.999$) across a wide range (up to 8 Gy), with a deviation from ideal linearity below 1% (section 3.1). The current signal showed negligible dependence on dose rate over the investigated range (0.1–3 Gy/min), confirming the stability of carrier collection even under variable irradiation conditions. Furthermore, the measured percentage depth dose (PDD) curves were consistent with those obtained using reference ionization chambers, with deviations within $\pm 3\%$ at all depths (section 3.2), and the field size dependence was modest ($< 5\%$) for diameters between 2 and 10 cm (section 3.3). The detector also showed excellent short-term reproducibility (fluctuations $< 0.5\%$) and long-term stability, with signal drift $< 1\%$ over several days of repeated irradiation (section 3.4). These results confirm and extend previous findings on

the robustness of SiC detectors in different radiation fields, including γ -rays, electron and proton beams [5, 7, 10, 11]. The superior radiation hardness, combined with low leakage current and reduced energy dependence compared to silicon, make this device particularly well-suited for use in preclinical QA procedures and in-vitro radiobiological experiments. Importantly, the encapsulated SiC detector proved compatible with immersion in water-equivalent phantoms, expanding its utility in depth-dose mapping. Future developments will include the characterization of these detectors under conventional clinical radiotherapy conditions, specifically with photon beams in the 6–15 MV energy range and electron beams typically between 6 and 20 MeV. Efforts will also focus on exploring their integration into clinical practice. In parallel, further miniaturization and advanced microfabrication techniques could enable the embedding of SiC dosimeters directly into small animal models or close to target volumes, allowing for highly localized in vivo dose verification.

Acknowledgments

This research was supported by the DOTSiC project (Telescopic Dosimeter in Silicon Carbide), funded by the National Scientific Committee for Technology Transfer (CNTT) of the Italian National Institute for Nuclear Physics (INFN). Additional financial support was provided by the Italian Ministry of University and Research (MUR) under the framework of the PRIN project titled Proton Boron Capture Therapy (PBCT).

References

- [1] M.M. Aspradakis et al., *SSRMP Recommendations No 9: Reference dosimetry in low and medium energy x-ray beams*, *Z. Med. Phys.* **33** (2023) 601.
- [2] R. Hill et al., *Advances in kilovoltage x-ray beam dosimetry*, *Phys. Med. Biol.* **59** (2014) R183.
- [3] M.A. Hill et al., *The importance of standardization and challenges of dosimetry in conventional preclinical radiation biology research*, *Br. J. Radiol.* **98** (2025) 993.
- [4] M. Bruzzi et al., *Advanced materials in radiation dosimetry*, *Nucl. Instrum. Meth. A* **485** (2002) 172.
- [5] I. Lopez Paz et al., *Position-resolved charge collection of silicon carbide detectors with an epitaxially-grown graphene layer*, *Sci. Rep.* **14** (2024) 10376.
- [6] S. Tudisco et al., *SiCILIA — Silicon Carbide Detectors for Intense Luminosity Investigations and Applications*, *Sensors* **18** (2018) 2289.
- [7] G. Petringa et al., *First characterization of a new Silicon Carbide detector for dosimetric applications*, *2020 JINST* **15** C05023.
- [8] F. Romano et al., *First Characterization of Novel Silicon Carbide Detectors with Ultra-High Dose Rate Electron Beams for FLASH Radiotherapy*, *Appl. Sci.* **13** (2023) 2986.
- [9] G. Milluzzo et al., *Comprehensive dosimetric characterization of novel silicon carbide detectors with UHDR electron beams for FLASH radiotherapy*, *Med. Phys.* **51** (2024) 6390.
- [10] M. Bruzzi and E. Verroi, *Epitaxial SiC Dosimeters and Flux Monitoring Detectors for Proton Therapy Beams*, *Materials* **16** (2023) 3643.
- [11] M. Bruzzi, F. Nava, S. Pini and S. Russo, *High quality SiC applications in radiation dosimetry*, *Appl. Surf. Sci.* **184** (2001) 425.

- [12] M. Bruzzi et al., *Characterisation of silicon carbide detectors response to electron and photon irradiation*, *Diam. Relat. Mater.* **10** (2001) 657.
- [13] PTW-Freiburg, *Farmer ionization chambers 30010*,
<https://www.ptwdosimetry.com/en/products/farmer-ionization-chambers-30010>.
- [14] PTW-Freiburg, *Advanced markus electron chamber — model 34045*,
<https://www.ptwdosimetry.com/en/products/advanced-markus-electron-chamber>.
- [15] Tektronix, *Keithley electrometers: Low-level, sensitive, and specialty instruments*,
<https://www.tek.com/en/products/keithley/low-level-sensitive-and-specialty-instruments>.